SI 385 | Revised Decision Memo

To: Credit Union Lending Committee | From: Marielle Johnson
Subject: Recommended Mileage Adjustment for Used Vehicles

1. The single analytical decision I would defend

I recommend Varga's adjustment of approximately $630 less in asking price per additional 10,000 miles as a starting point for the credit union. The main analytical decision I defend is controlling for vehicle age instead of using mileage alone. In DoU02_varga.py, Varga excludes listings priced below $500 or with odometer readings outside 100 to 500,000 miles, then regresses price on mileage and age. My DoU02_mj.py cell, 'Comparing the Regression Results,' reproduces his coefficient of -$633.06 using 3,645 listings, with an R-squared of 0.267. Age matters because two cars with the same mileage may have different asking prices if one is much older. I cannot claim that age alone causes the difference from Mensah's result, since Varga also changes the sample. Even with that limitation, I prefer his approach because it accounts for an important vehicle characteristic without requiring a highly detailed model. This is a provisional recommendation, not a final lending rule.

2. The alternative I rejected, and why

Mensah's DoU02_mensah.py offers a strong alternative. Its mileage-only OLS regression is simple to explain and uses more listings without Varga's additional filtering rules. My comparison cell reproduces -$277.06 per 10,000 miles from 4,086 listings, with an R-squared of 0.053. That estimate could be useful as a broad description of the listings, but it does not separate mileage from vehicle age. For a lending adjustment, I find that limitation important. I also considered my own manufacturer model in DoU02_mj.py. The 'My Regression: Including Vehicle Manufacturer' cell gives -$757.39 using 3,529 listings, with an R-squared of 0.438. Yet my 'Checking the Effect of Manufacturer' cell shows that Varga's formula on those same 3,529 listings gives -$719.74. Changing the sample accounts for an $86.68 shift from Varga's original coefficient; adding manufacturer accounts for another $37.65. Therefore, I reject my larger estimate as the policy figure: a higher R-squared does not settle the question, especially when the sample also changes.

3. What I am still unsure about

I am still unsure whether any single mileage adjustment is stable enough for actual lending decisions. My same-sample check changed Varga's estimate from -$633.06 to -$719.74 without changing the regression formula. I have not established exactly which excluded listings drive that shift. I also cannot tell whether Varga's cutoffs remove legitimate inexpensive or high-mileage cars, potentially making the sample less representative of the credit union's borrowers. Finally, these are Craigslist asking prices, not completed sale prices. Before using $630 as a fixed adjustment, I would test other reasonable samples and compare the results with actual transaction data.

Revision account | approximately 100 words

Clear change from first draft: I changed the $630 recommendation from a mainly age-based argument to a provisional recommendation that explicitly addresses sample sensitivity.

I used AI to draft the first memo, which I have attached separately. My most important revision was changing how strongly I defended Varga's $630 estimate. The first draft focused on controlling for age and mentioned sample sensitivity, but it did not clearly connect the recommendation to all three code versions. In the revised memo, I named Mensah's, Varga's, and my notebooks and added my same-sample findings: -$633.06 became -$719.74 without changing Varga's formula, then -$757.39 after adding manufacturer. That evidence made me present $630 as a provisional recommendation, rather than a settled lending rule. I also explained the trade-off more explicitly.

AI disclosure: AI assisted with the first draft and this revision. The original AI-generated first draft is supplied separately.